SAP CPI — Class Notes

Day 1: Introduction to SAP & SAP CPI

Typed and lightly expanded version of handwritten class notes
Dated 24 Aug 2026

Day 1 — Introduction to SAP CPI (24 Aug 2026)

1. What Is SAP?

- **SAP** should be pronounced letter-by-letter: **S-A-P** (not "sap" as one word)
- **Full form:** Systems, Applications, and Products (in Data Processing)
- **Founded:** 1972
- **Headquarters:** Germany (Walldorf)

2. The Core Business Problem — Grocery Shop Example

To understand why SAP exists, start with a simple example: a **grocery shop**.

Running a single shop requires managing four key areas:

Area	What It Covers
Materials	Stock/goods available in the shop
Place	Physical location/store
Financial	Money, billing, accounts
Sales	Actual selling of goods to customers

For a single small shop in a village, just **1–2 people** can manage all of this manually.

But scale this up: imagine the same business now has shops across **multiple cities**, needing **200–300 employees** to run it. At this scale, manual tracking breaks down — and this creates a real risk: **chance of fraud occurring from any point/action in the business**, simply because no one has a complete, real-time view of everything happening across all locations.

3. The Solution: ERP

To solve this exact problem, **SAP (the company)** introduced a software category called **ERP**.

> **ERP = Enterprise Resource Planning**
> A single software that runs all the required resources of a business on one unified platform.

4. SAP Modules: Functional vs Technical

SAP is organized into different modules, split into two broad categories:

Functional Modules	Technical Modules
SAP MM (Materials Management)	SAP ABAP
SAP FICO (Finance & Controlling)	**SAP CPI** (our focus)
SAP SD (Sales & Distribution)	
SAP HR (Human Resources)	

Rule of thumb: Functional modules map to specific business departments (materials, finance, sales, HR). Technical modules (ABAP, CPI) are about building and connecting the systems that run underneath those business functions.

5. Why SAP Is Useful — The D-Mart Example

Scenario: A customer walks into D-Mart wanting to buy **10 bags of rice**.

Problem: The 10 bags are not available.

Result: Business loss.

Here's why this happens without a proper system — trace the process flow:

Customer	→	Sales	→	Inventory	→	PP (Production Planning)	→	Delivery

The customer has to wait through this entire chain — this is called the **"Total Waiting Time."** If any link in this chain is slow, disconnected, or outdated, the result is the same: **the business ultimately loses the sale.**

6. The Fix: SAP ERP System (Centralized System)

To overcome this issue, SAP introduced the **SAP ERP System** — a **Centralized System**.

Instead of Sales, Inventory, Production, and Delivery operating as disconnected silos, they all connect to and communicate through **one central ERP System**:

Centralized ERP System				
Sales (S)	↔	**ERP System**	↔	Materials Mgmt (MM)

↕ (also connects to Production, Purchasing, and other modules)

Worked Example — Real-Time Visibility

Say the availability of rice bags is tracked in the centralized ERP system:

- Morning stock: **100 bags**
- By end of day: **95 bags sold**
- Remaining: **5 bags** — and this number is instantly visible to **every single person/department** connected to the system

Outcome: Because everyone sees the same real-time number, the business will **never end up with an empty-stock surprise** — sales, purchasing, and production can all react in time.

7. Core Functional Modules — What Each One Actually Does

Module	Full Form	What It Handles
SAP MM	Materials Management	Raw materials, stock availability checking
SAP FICO	Finance & Controlling	Processing finance, profit, loss, investment
SAP SD	Sales & Distribution	Orders, delivery, billing
SAP PP	Production Planning	When to start preparing/producing goods, etc.
SAP HR *(SuccessFactors)*	Human Resources	Employee details, pay details, leaves, etc.

8. Real Companies Using SAP

Examples of companies using SAP services: **Reliance, Adani, Hero,** and others — spanning industries such as:

- Manufacturing
- Pharmaceutical
- Retail and Marketing
- Gas companies (HP, Bharat, etc.)

All of these are grouped under the umbrella term: **Business Applications.**

9. Now, About CPI

CPI = Cloud Platform Integration

This is a **technical module**. Its core job: transferring data between two or more systems.

Real Example — Honda & MRF

Honda is a manufacturing company. Honda needs **tires** to complete/accomplish its final product (a vehicle). So, Honda needs to get tires from **MRF** (a tire company).

This means **data has to transfer between these two companies' systems** — and this data transfer/integration is handled **by SAP** (specifically, by SAP CPI).

10. The Three Types of Data Flow CPI Handles

SAP	↔	SAP
SAP	↔	non-SAP

> **SAP CPI is a Middleware** — it is the connector layer that sits between systems and manages this data flow.

11. Other Middleware Services (CPI's Competitors)

- **MuleSoft** — owned by Salesforce
- **TIBCO**
- **Boomi** *(previously "Dell Boomi" — Boomi was later sold by Dell and is now an independent company)*

12. Why SAP CPI Is In Demand

SAP CPI is currently in strong demand because:

- It is **SAP's own product** — built specifically for the SAP ecosystem
- It is a **cloud service** — aligned with where the industry is heading
- **SAP is the market leader** in the ERP space, which drives natural demand for its native integration tool

> **Note (added during review):** SuccessFactors is technically SAP's *cloud-based* HR product, while "SAP HR/HCM" refers to the older on-premise HR module. Most companies today use SuccessFactors — the class notes group them together, which is fine conceptually for Day 1, but worth knowing the distinction for later or for interviews.